Armenian IT Curriculum vs. Labor Market Alignment

Master's thesis progress presentation

Research Question

- How well do IT curricula from selected Armenian universities align with current Armenian IT job market demands?
- Goal: build a reproducible, data-driven pipeline instead of relying only on qualitative discussion.
- Current stage: analytical pipeline complete, thesis writing and coverage improvement in progress.

Why This Matters

4 Universities
current curriculum datas

753 IT-only jobs
analysis subset after filtering

Practical value

- Shows where curricula already match market demand.
- Identifies gaps in applied competences and modern tools.
- Can later be reused for repeated alignment checks.

Project Scope and Current Dataset

Education corpus and labor-market corpus used in the current analysis

1,161 Curriculum Data
courses from 4 universities / 25 programs

1,369 Broad Job Snapshot
postings from 14 sources

753 IT-Only Set
postings kept for downstream NLP

University	Programs	Courses	Description availability
YSU	13	691	Translated Armenian description
AUA	7	249	English descriptions
NUACA	4	174	Mostly names only
RAU	1	47	Mostly names only

Important Scope Note

- This is not yet the full Armenian higher-education landscape.
- NPUA and UFAR are still missing from the dataset.
- RAU is only partially covered at this stage.

How University Data Was Collected

Different universities required different collection methods because their data formats were very different

YSU

Apify + rendered markdown pages

691 courses I Armenian I rich text after translation

AUA

direct HTML course catalog parsing

249 courses I English I strongest description coverage

NUACA

web scraping of plain-text course lists

174 courses I English names only

RAU

PDF parsing with PyPDF2 + regex

47 courses I Russian -> English titles

Key takeaway: the curriculum corpus was built from heterogeneous web pages and PDFs, then merged into one unified dataset.

How Job Data Was Collected

The job corpus combines broad aggregators with direct employer portals

Main public sources

LinkedIn | Staff.am | job.am

Company portals used in the current snapshot

EPAM | SoftConstruct | Grid Dynamics | Krisp | NVIDIA | 10Web | DataArt | ServiceTitan | Synopsys | Picsart | DISQO

Scraping / collection methods

- LinkedIn: Apify cloud scraper
- Staff.am and job.am: requests + BeautifulSoup
- EPAM: internal search API
- Picsart / DISQO: public ATS APIs
- DataArt / ServiceTitan: Playwright for dynamic content
- SoftConstruct / Krisp / Synopsys: server-rendered HTML parsing

Broad snapshot result: 1,369 postings from 14 sources

Cleaning, Structuring, and IT-Only Filtering

The broad market scrape had to be narrowed before downstream NLP analysis

IT-only funnel

What happened in this step

- Merged multiple raw schemas into one canonical dataset.
- Built a full_text field for downstream text analysis.
- Removed repeated boilerplate from job descriptions.
- Created a keep / drop / review audit for IT-only filtering.
- Preserved the broad market snapshot for transparency, but used the IT-only subset for analysis.

Why this matters

Without this step, non-IT and mixed-role postings would distort the demand-side skill signal.

Translation and Multilingual Handling

YSU curriculum content had to be made comparable to the mostly English job corpus

Why translation was needed

- YSU data is in Armenian, while most job postings are in English.
- A shared English pipeline is easier to validate and explain than a mixed-language pipeline.
- Translated text can still be inspected manually by a reviewer.

Provider comparison

Provider	Manual validation score
OpenAI gpt-4o-mini	20 / 20
Perplexity Sonar Pro	6 / 20

Chosen solution

- Translate Armenian curriculum text to English first.
- Keep technical terms preserved during translation.
- Use the translated output as the main NLP input.

EDA and Data Understanding

Exploratory analysis was used to understand asymmetry and corpus quality before NLP

Curriculum-side asymmetry

Description availability drives concept richness.

AUA and YSU have much richer text for NLP than NUACA and RAU.

Job-side structure

Main EDA finding

- LinkedIn dominates the IT-only market subset.
- Company portals still add important employer-specific signal.
- The demand corpus is not uniform and must be interpreted carefully.

TF-IDF Extraction: Frequency-Based Baseline

Used as the main transparent baseline for candidate skill extraction

What TF-IDF means

TF-IDF(term, document) = term frequency x inverse document frequency

- Finds words or short phrases that are important inside one document relative to the full corpus.
- Transparent and easy to explain.
- Worked better than KeyBERT in validation against human skill tags.

Why it was used here

- Strong baseline for technical vocabulary.
- Good at recovering repeated shared terms such as Python, SQL, cloud, testing.
- Useful for comparing curriculum and market vocabulary before normalization.

Known weakness

Can produce awkward fragments and generic terms if preprocessing is weak.

TF-IDF Example on My Data

Example from the course: Information Technologies in the Professional Field (Python)

Input document

Application of the fundamentals of the Python programming language, working with data, variables, arrays, functions...

->

Top TF-IDF outputs

- python
- data
- language data variables
- arrays functions
- language data

Takeaway

TF-IDF captured the main topic well, especially the core technology word 'python'.

KeyBERT Extraction: Semantic Phrase Method

Used as a semantic comparison method against TF-IDF

What KeyBERT means

Score(candidate phrase) = cosine similarity(document embedding, phrase embed

- Uses embeddings to find phrases that are semantically close to the whole document.
- Produces richer multi-word phrases than TF-IDF.
- Useful for testing whether semantic extraction improves the alignment story.

Why it matters in this thesis

- Captures meaning, not only frequency.
- Can recover more descriptive keyphrases from short course text.
- But raw overlap stays very low because these phrases rarely match job phrases exactly.

KeyBERT Example on My Data

Same course example used to compare semantic extraction with TF-IDF

Input document

Application of the fundamentals of the Python programming language, working with data, variables, arrays, functions...

->

Top KeyBERT outputs

- data visualization python
- visualization python teaching
- fundamentals python
- python teaching
- teaching create visualizations

Takeaway

KeyBERT gives richer phrases, but these phrases usually do not match job phrases di

Validation, Sensitivity Analysis, and Noise Audit

The NLP outputs were tested instead of being treated as automatic ground truth

1. Description asymmetry

AUA setup	Coverage
Names only	1.3%
Names + descriptions	6.8%

Result: richer descriptions produce about a 5x multiplier.

2. Human skill-tag validation

Metric	TF-IDF	KeyBERT
Soft recall	44%	21%
Jobs with >=1 tag matched	84.2%	38.4%

Result: TF-IDF clearly outperformed KeyBERT on this validation

3. Noise audit

Initial TF-IDF overlap

584 terms (12.6%)

After stronger filtering + IT-only scoping

279 terms (8.85%)

Result: many apparent overlaps were generic English words, not re

ESCO Normalization: The Taxonomy Bridge

This step turns different surface forms into comparable shared skill concepts

How it works

- Encode extracted phrases and ESCO labels with the same embedding model.
- Compare them by cosine similarity.
- Keep matches above the calibrated threshold.
- Use ESCO concepts as the common comparison layer between curricula and jobs.

Threshold calibration

Item	Value
Calibration pairs	293
Selected threshold	0.75
Best F1	0.711

Example mapping

Raw phrase	ESCO concept
python programming	Python (programming language)
python development	Python (programming language)
object oriented programming	object-oriented programming
OOP principles	object-oriented programming

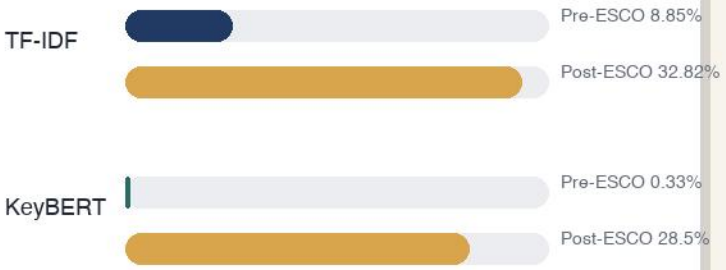
Key message

This step is what turns vocabulary overlap into concept overlap.

Before and After ESCO Normalization

Concept-level alignment is much stronger than raw string overlap

Coverage comparison



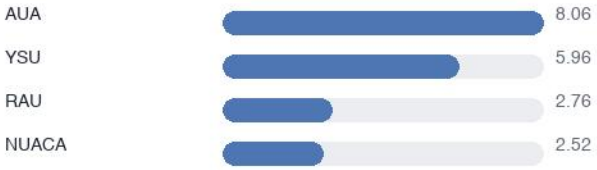
Interpretation

- Raw string overlap severely underestimates real conceptual overlap.
- TF-IDF remains the stronger main method in this project.
- KeyBERT becomes more useful only after normalization.
- The ESCO-normalized metric is still a lower bound because ESCO misses many modern tools.

Current Alignment Findings

What the current pipeline shows about Armenian curricula and employer demand

Average coverage by university



Best program: AUA Computer and Information Science (Master) - 12.27%

Lowest current program: NUACA Geographic Information Systems (Master) - 0.92%

Top demanded skills in the IT-only market

Main finding

Curricula overlap with the market more at the knowledge level than at the applied tool /

Limitations and What I Will Do Next

Tomorrow's presentation should show both the progress and the remaining gaps honestly

Main limitations

- NPUA is still missing from the dataset.
- UFAR is still missing from the dataset.
- RAU is only partially covered.
- Description asymmetry remains strong across universities.
- ESCO misses modern tools such as Docker, React, Azure, Kubernetes, and CI/CD.

Immediate next steps

- Try to get in touch with UFAR for curriculum data.
- Try to get in touch with NPUA for curriculum data.
- Try to contact or visit RAU to improve coverage beyond one program.
- Continue refining the thesis draft and interpretation chapters.

How to frame this tomorrow

The pipeline is complete and working; the next phase is improving data coverage and f